

L9a: Introduction to Derivatives and European Option Pricing

CHEME 5660 Cornell University

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Objectives for Today

A Kalshi contract costs less than a dollar today and pays one dollar only if an event occurs. What is a payment that may never arrive worth today?

Today's concepts

- Define and classify derivatives: futures, forwards, swaps, and options.
- Analyze option payoffs and profits: long and short calls and puts at expiration, breakevens, and loss bounds.
- Price European options contracts: the premium makes the expected net present value zero under the risk-neutral measure, the Black–Scholes–Merton formulas compute it, and put–call parity links calls and puts.

Lectures and Examples

Lecture:
CHEME-5660-L9a-Lecture-IntroductionToDerivatives-BSM-Fall-2026.ipynb

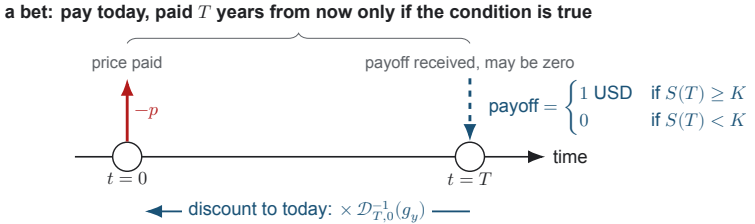
Example 1:
CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:
CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

The first example computes payoff and profit for single long and short call and put contracts. The second computes European premiums with Black–Scholes–Merton and Monte Carlo simulation, then varies the strike.

Concept Review: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if $S(T) \geq K$, and nothing otherwise.



Discounting the payoff at the benchmark growth rate g_y gives a random NPV:

$$\text{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S(T)) = \begin{cases} -p + e^{-g_y T} & \text{if } S(T) \geq K \text{ (true)} \\ -p & \text{if } S(T) < K \text{ (false)} \end{cases}$$

Expected NPV of the Bet

Under our beliefs about $S(T)$, the expected NPV is:

$$\mathbb{E}[\text{NPV}_0] = (-p + e^{-g_y T}) \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} + (-p) \underbrace{\mathbb{P}(S(T) < K)}_{\text{false}} \qquad \text{weight by probability}$$

$$= -p \underbrace{[\mathbb{P}(S(T) \geq K) + \mathbb{P}(S(T) < K)]}_{=1} + e^{-g_y T} \mathbb{P}(S(T) \geq K) \qquad \text{collect the } p \text{ terms}$$

$$= -p + e^{-g_y T} \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} \qquad \text{probabilities sum to one}$$

The bet has a non-negative expected NPV when $p \leq e^{-g_y T} \mathbb{P}(S(T) \geq K)$.

The bet is a derivative contract: its payoff depends on the future price of another asset. Every buyer pays one market price, yet buyers with different beliefs value the same contract differently.

How is the market price set?

Company Profile: Cboe Global Markets

Cboe opened on April 26, 1973, as the Chicago Board Options Exchange, the world's first exchange for listed options.

- Listed options: standard strikes, expirations, and contract sizes, cleared by the Options Clearing Corporation (OCC).
- Index options and volatility: SPX options are European-style and cash-settled. The VIX reads the market's expected 30-day volatility of the S&P 500 from their premiums.
- Event contracts: since June 2026, binary options on the Mini-S&P 500 that pay 100 USD or nothing.

Today's question: where does a listed premium come from?

What Is a Derivative?

A derivative is a contract between a buyer and a seller to transact an asset at a set price on a future date. It is an abstract asset whose cash flows depend on an underlying asset or event.

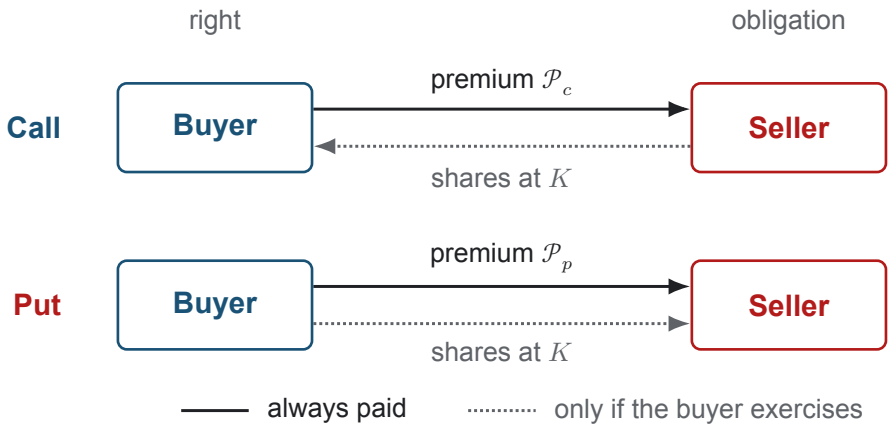
- Futures, forwards, and swaps: obligate both parties to transact. Futures trade on exchanges with standard terms. Forwards trade over the counter, where each party bears the other's default risk. Swaps exchange one asset's cash flows for another's, also over the counter.
- Options: the buyer holds the right, but not the obligation, to buy (call) or sell (put) at a set price. Options are exchange-traded.
- Event contracts: pay a fixed amount, usually one dollar, if an event occurs and nothing otherwise. Kalshi lists them.

How Big Is the US Options Market?

- 2025: a record of about 15.2 billion contracts, or about 60.6 million per day (OCC).
- 2026: through September, about 13.2 billion contracts, or about 70.6 million per day (OCC).
- Notional: about 4 trillion USD per day in 2025, more than three times the 1.1 trillion USD per day traded in U.S. cash equities (Traders Magazine, Cboe).

Sources: the lecture notebook's market table, with dates and links.

Call and Put Options Contracts



The Business Case for Long and Short Calls

A call option gives the owner the right, but not the obligation, to buy 100 shares of an underlying asset (per contract) at the strike price K by the expiration date.

- Buyer (long): benefits from upside price movement without purchasing the asset directly. Long calls have defined risk: the maximum loss is the premium paid. If the owner exercises, they buy shares at K instead of the market price.
- Seller (short): receives the premium. Short calls have unlimited upside risk, so calls are often sold only by investors who already own the shares, as a covered call (100 shares per contract). Short calls obligate the seller to sell shares at K if the option is exercised.

The Business Case for Long and Short Puts

A put option gives the owner the right, but not the obligation, to sell 100 shares at the strike price K by the expiration date.

- Buyer (long): benefits from downside price movement without selling the asset directly. Long puts have defined risk: the maximum loss is the premium paid. The owner can sell shares at K instead of the market price.
- Seller (short): receives the premium. Short puts have large but bounded downside risk: if the share price falls to zero, the seller loses the strike price minus the premium's future value per share. Puts are often sold only by investors with the cash to buy the shares (100 per contract), a cash-secured put. Short puts obligate the seller to buy shares at K if the option is exercised.

Call and Put Payoffs

The payoff per share of a single call or put contract at expiration (or when the contract is exercised) is:

$$\boxed{\begin{aligned} V_c(K, S(T)) &= \max(S(T) - K,\ 0) = (S(T) - K)^+ \\ V_p(K, S(T)) &= \max(K - S(T),\ 0) = (K - S(T))^+ \end{aligned}}$$

where K is the strike price and $S(T)$ is the share price at expiration (time $t = T$). The seller charges the buyer a premium per share, $\mathcal{P}_c$ for a call and $\mathcal{P}_p$ for a put.

Profit at Expiration

The premium is paid at $t = 0$ and the payoff arrives at $t = T$, so we compare them at expiration using the premium's future value, $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$, where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$. The buyer's profit per share for a single contract is:

$$P_c^{\text{buyer}}(K, S(T)) = V_c(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$$
$$P_p^{\text{buyer}}(K, S(T)) = V_p(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$$

A position's orientation θ is $+1$ for the buyer (long) and -1 for the seller (short). Its profit per share is $\theta \cdot P^{\text{buyer}}$, so the seller's profit is $P^{\text{seller}} = (-1) \cdot P^{\text{buyer}}$, and $P^{\text{buyer}} + P^{\text{seller}} = 0$ at every share price.

Options are a zero-sum game: one party's gain is the other's loss.

Breakeven

Setting the buyer's profit to zero gives the breakeven share price at expiration:

$$S_{\text{BE}} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c \qquad \text{call}$$

$$S_{\text{BE}} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p \qquad \text{put}$$

Above the call breakeven, and below the put breakeven, the buyer profits and the seller loses. Listed contracts often expire within weeks, so $\mathcal{D}_{T,0}(g_y)$ is close to one (about 1.003 for 30 days at $g_y = 4\%$).

Example 1: CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

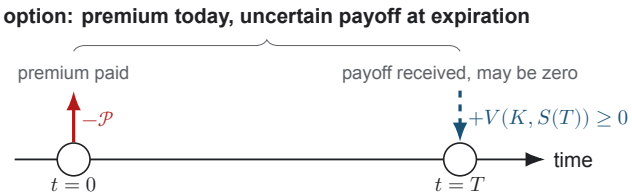
Exercise Styles

What is an option's payoff worth today? The answer depends on when the holder can exercise:

- European: exercise only on the expiration date.
- American: exercise at any time before or on the expiration date. This creates extra risk for the seller, so American-style options cost at least as much as otherwise identical European-style options.
- Bermudan: a less common hybrid, exercised only on specified dates up to the expiration date.

Let's start with European-style options and compute their net present value.

Options as Abstract Assets



The buyer pays the premium $\mathcal{P}$ today and receives the payoff $V(K, S(T)) \geq 0$ at expiration, so the net present value at $t = 0$ is:

$$\text{NPV}_0 = -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T))$$

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ gives $\text{NPV}_0 = \mathcal{D}_{T,0}^{-1}(g_y) \cdot P^{\text{buyer}}$: the NPV is the buyer's profit at expiration discounted to today, so it is zero at the breakeven share price.

The European Premium

A Treasury's cash flows are known, so its price makes the NPV zero. An option's payoff is random. Suppose a replicating portfolio of shares and the risk-free asset pays the option's payoff in every outcome. With no transaction costs, any premium other than its cost creates a risk-free profit:

- Too low: traders buy the option and sell the portfolio.
- Too high: traders sell the option and buy the portfolio.
- At cost: neither trade earns a risk-free profit.

Under the risk-neutral measure $\mathbb{Q}$, the share's expected price grows at g_y and the option's expected NPV is zero:

$$\mathbb{E}_{\mathbb{Q}}(\mathrm{NPV}_0) = 0 \implies \boxed{\mathcal{P}^{\mathsf{EU}} = \mathbb{E}_{\mathbb{Q}}(\mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T)))}$$

What $\mathbb{Q}$ Means

$\mathbb{Q}$ is a pricing rule, while the real-world probabilities $\mathbb{P}$ still describe how often outcomes occur:

- The share: with no dividends, $\mathbb{E}_{\mathbb{Q}}(\mathcal{D}_{T,0}^{-1}(g_y) \cdot S(T)) = S(0)$, so the share also has zero expected NPV under $\mathbb{Q}$.
- The option: under $\mathbb{P}$, a call expires worthless with probability $\mathbb{P}(S(T) \leq K)$, from the terminal target formula, and the buyer then loses the whole premium.
- The event contract: a one-dollar contract costs $p = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{Q}(\text{event})$. The pricing probability $\mathbb{Q}(\text{event})$ can differ from $\mathbb{P}(\text{event})$.

The Concept Review bet is the event contract on $S(T) \geq K$, so $p = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{Q}(S(T) \geq K)$. A general Kalshi event has no replicating portfolio, so choosing its pricing probability requires additional assumptions.

Why Is the Bet So Cheap?

Take a one-year bet that a share finishes at or above today's price, with 30% volatility, a 6% real-world mean growth rate, and a 4% risk-free rate:

- Real world: the bet pays off 58% of the time.
- A client with 50,000 USD and relative risk aversion 3 would pay 0.557 USD for it.
- The market charges only 0.474 USD.

Where did the other 8 cents go? The answer depends on when the bet pays, not on how risky the bet is on its own.

Optional: contingent claims notebook, last section.

The Black–Scholes–Merton Equation

Assume GBM with constant σ and g_y, no dividends, and frictionless continuous trading. Write the premium at time t, when the share price is S, as $\mathcal{P}(S, t)$, with $\mathcal{P}(S, T) = V(K, S)$. It satisfies:

$$\boxed{\frac{\partial \mathcal{P}}{\partial t} + g_y S \frac{\partial \mathcal{P}}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \mathcal{P}}{\partial S^2} = g_y \mathcal{P}}$$

- Hedge: hold one option and sell $\Delta = \partial \mathcal{P}/\partial S$ shares. The random changes cancel, so the portfolio is risk-free.
- No arbitrage: a risk-free portfolio grows at g_y, which gives the equation. The real-world expected growth cancels.
- Solution: solving backward from the payoff gives $\mathcal{P}(S(0), 0) = \mathcal{P}^{\mathsf{EU}}$.

Derivation: the contingent claims notebook.

Black–Scholes–Merton Premiums

To compute the expectation under $\mathbb{Q}$, we can simulate GBM with drift g_y, or use the closed-form European call and put premiums:

$$\boxed{\begin{aligned}\mathcal{P}_c &= N(d_+)\cdot S(0) - N(d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y)\\ \mathcal{P}_p &= N(-d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y) - N(-d_+)\cdot S(0)\end{aligned}}$$

where N is the standard normal CDF, T is in years, and:

$$d_+ = \frac{1}{\sigma\sqrt{T}}\left[\ln\frac{S(0)}{K} + \left(g_y + \frac{\sigma^2}{2}\right)T\right], \qquad d_- = d_+ - \sigma\sqrt{T}$$

Merton and Scholes received the 1997 Nobel Prize in Economic Sciences. Fischer Black had passed away before the prize was awarded.

Put–Call Parity

For a call and a put with the same strike and expiration, the payoffs differ by $S(T) - K$ at every share price:

$$V_c(K, S(T)) - V_p(K, S(T)) = (S(T) - K)^+ - (K - S(T))^+ = S(T) - K$$

Subtracting the put premium condition from the call premium condition, and using $\mathbb{E}_{\mathbb{Q}}(\mathcal{D}_{T,0}^{-1}(g_y) \cdot S(T)) = S(0)$, gives:

$$\boxed{\mathcal{P}_c - \mathcal{P}_p = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y)}$$

Parity uses only the payoff identity and the share's zero expected NPV, so it holds whatever model describes the share price under $\mathbb{Q}$.

Example 2: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

What About American-Style Options?

The holder can exercise at any time $\tau \leq T$, chosen using only the information available at that time. The premium is the largest expected payoff, discounted from τ back to today, over all such exercise times:

$$\mathcal{P}^{\mathsf{AM}} = \sup_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\Big) \geq \mathcal{P}^{\mathsf{EU}}$$

The American put has no closed-form formula like BSM, so its premium is usually computed on a binomial lattice.

Optional Advanced Material

Contingent claims: advanced/contingent_claims/
CHEME-5660-L9a-Advanced-ContingentClaims-BSM-Fall-2026.ipynb

Where do the BSM premiums come from? Replication on a one-step lattice gives the risk-neutral probability and the state prices that price event contracts. A continuous hedge gives the BSM equation, whose solution is the risk-neutral expectation. The call is an asset-or-nothing claim minus K digital claims. Marginal utility then shows where the bet's risk premium goes.

Volatility skew: advanced/spxw_volatility_skew/
CHEME-5660-L9a-Advanced-SPXW-Volatility-Skew-Fall-2026.ipynb

Does one volatility price every strike? We invert a synthetic SPXW-style chain of European, cash-settled index options for implied volatility.

Notebooks: lectures/week-9/L9a/advanced, also linked from the lecture notebook.

Summary

Today we introduced derivatives, read the payoff and profit of call and put contracts, and priced European contracts.

Key takeaways

- Derivatives derive their value from underlying assets: common types are futures, forwards, swaps, and options.
- Option payoffs are set by the strike and the position: the premium turns payoff into profit and sets the breakeven.
- The premium makes the expected NPV zero under $\mathbb{Q}$: the BSM formulas compute it for European contracts, and parity links calls and puts.